Polypharmacy Adverse Drug Event (ADE) Cohort Pipeline (ed_non_opioid)

Phase 0: Virginia APCD Longitudinal Lake (2016–2020)
Medical Claims Lake
(All-Payer Encounters, HCG Detail Codes, ICD-10 Positions 1–10)
Pharmacy Claims Lake
(Adjudicated Dispenses, NDC to Generic RxNorm, Days Supply)
Annual Partitions (2016, 2017, 2018, 2019)
Stratified into 8 Age Bands (0–12 to 85–114)
Claims Stream
Enrollment Roster
Phase 1: Case Anchor & Opioid Cleanse (phase1_data_preparation.py)
Milliman HCG Clinical Emergency Anchor
Target Codes: P51b (ER), O11 (ER Services), P33 (Outpatient Observation)
Target Timestamp: first_ed_non_opioid_date (t_0)
Raw ED Candidates
Absolute Opioid Cleanse Boundary
Strictly drops any patient with ANY opioid ICD-10 code
(F11.* or T40.*) across all 10 diagnosis columns in study year
Guarantees 100% mutual exclusivity with opioid cohort
Cleaned ED Cases
Phase 2: Sequential ADE Filtering Funnel (phase2_event_processing.py)
Filter 1: ED Utilization Cap (< 7 ED Visits/Year)
• Excludes chronic crisis utilizers ('flyers')
• Removes severe socio-environmental crisis noise
• Isolates true acute medication-mediated emergencies
Pass Filter 1
Filter 2: Proximal Drug Exposure Window (1–21 Days Pre-ED)
• Patient must have active pharmacy dispensing 1–21 days prior
• FDA FAERS Pharmacokinetic Validation (qa_faers_time_to_onset.py):
21-day window captures ~90.5% of all acute ADE time-to-onset distributions
Pass Filter 2
Filter 3: Zero-Day Discharge-Fill Guard
• STRICTLY PURGES all prescriptions dispensed on Day 0 (t_0)
• Clinical Rationale: Day 0 claims represent discharge meds
(e.g., antiemetics, antibiotics, analgesics) prescribed AFTER the visit
• Prevents immediate reverse-causality target leakage
1 Target Case
Validated ADE Cases
Phase 4: 1:5 Matched Control Sampling (phase3_cohort_creation.py)
Eligible Control Pool
• Covered APCD members in same partition year
• Zero qualifying ED visits in study year
• Verified health system contact continuity
Candidate Pool
1:5 Incidence-Density Matching (No Replacement)
Exact Stratification On:
1. Age Band (8 Strata: 0–12, 13–24, 25–44, ..., 85–114)
2. Biological Sex (Exact M/F Match)
3. Calendar Year Partition (2016–2019)
4. Active Coverage Continuity
Assign Anchor
Pseudo-Index Reference Date (t_pseudo)
Controls assigned earliest non-ED medical claim date
Guarantees identical 21-day pre-index baseline window
5 Matched Controls
Phase 3: Acute Exposure Horizon & Feature Engineering
21–30 Day Acute Pre-Index Window
[t_0 - 21/30 days → t_0 - 1 day]
• Active Multi-Drug Concurrency & Drug-Drug Interactions
• Beers Criteria High-Risk Medications in Older Adults
• Anticholinergic & Sedative Cumulative Burden
• Sudden Titrations & Recent Regimen Modifications
Extract Whitelist
Toolbox Demarcation Architecture
• Tabular Counts & Claims-Density Bins (P25/P50/P95) → Tree Feature Matrix
• bupaR Process Flows, DTW Clusters & FP-Growth Rules → Dashboard Only
• Locked 682-Feature Pre-Index Whitelist Enforced
Tabular Feature Matrix
Phase 5: QA Assertion & Checkpointed Export (phase4_finalization.py)
QA Passed
Cell-Size Suppression Guard
Suppresses n ≤ 10 cell counts per CMS/VHI privacy policy
Write Gold Artifact
Final Parquet Fact Table (s3://pgx-repository/cohorts/)
• schema: final_cohort_schema.json
• Cases + 5:1 Controls with 682 Whitelisted Features
• Checkpointed to S3 pgx-pipeline-status/

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DuckDB QA Unit Assertion (qa_index_date_uniqueness.py)
SELECT mi_person_key, count(DISTINCT first_ed_non_opioid_date)
HAVING count > 1;
Asserts EXACTLY ONE index date per person-year
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